Advanced Financial Accounting

An IFRS® Standards Approach, 3e

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Solutions Manual

Chapter 8

Accounting for the Effects of Changes in Foreign Exchange Rates

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CHAPTER 8

CONCEPT QUESTIONS

1 In accounting exposure, items in the financial statements which are exposed to changes in foreign exchange rates are assets and liabilities which are remeasured or translated at the closing rate. Accounting exposure stems from transactions denominated in foreign currencies and from translation of the financial statements of foreign operations. Accounting exposure measures the effect of a change in the nominal foreign exchange rate and can be measured quantitatively. Operating exposure is due to a change in the real exchange rate between two currencies. Operating exposure affects a firm's competitive position through its impact on a firm's revenue and costs. Operating exposure cannot be measured or estimated reliably.

2 Transaction exposure arises from contractual obligations and rights as a result of a firm entering into a foreign currency denominated transaction. Translation exposure arises when the foreign currency financial statements of an overseas operation are remeasured or translated into the presentation currency.

3 A functional currency is the currency of the primary economic environment in which a firm operates. For a stand-alone entity, the designation of a currency as the functional currency means that all other currencies other than the functional currency are foreign currencies. For a group entity, the choice of functional currency of a foreign operation, such as an overseas subsidiary, determines the method of translating the financial statements of the foreign operation.

4 Under the Closing Rate method, the foreign operation is considered as an autonomous operation and the parent's interest in the foreign operation is mainly as an investment. As such, the interest of the parent is in the net assets of the foreign operation and its financial performance. Assets and liabilities of the foreign operation are translated at closing rate. Under the remeasurement approach, the foreign operation is considered to be an extension of the parent company's operation. Transactions of the foreign operation is deemed as transactions undertaken by the parent and the assets and liabilities of the foreign operation are deemed as assets and liabilities of the parent. Monetary items are remeasured at closing rate and non-monetary item carried at cost are remeasured at historical exchange rates.

5 Translation differences under the Closing Rate method do not have an impact on the parent's cash flow or operations and hence are taken to equity. The accumulated amount in the foreign currency translation reserves are released to profit or loss when the foreign operation is disposed of. Under remeasurement, remeasurement gains or losses are taken to the income statement as they are considered to be 'transaction exchange' differences.

6 A change in the functional currency is accounted for prospectively.

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7 Yes. Goodwill is considered an asset of the foreign operation and is translated using the closing rate under the Closing Rate method.

8 Answer is (c). The functional currency of the subsidiary in England is a third currency. This requires the financial statements prepared in British pounds to be remeasured into the functional currency, the US dollar, and then translated using the closing rate method into the presentation currency.

EXERCISES

Exercise 8.1
Answer is (b).

	FC000	Rate	$000
Revenue	1,000	0.6	600
Depreciation expense	(200)	0.5	(100)
Remeasurement gain*			100
Net profit	800		$600

Remeasurement gain:

	FC000	Rate	$000
Opening net monetary asset	2,000	0.5	1,000
Revenue	1,000	0.6	600
Purchase of cars	(2,000)	0.5	(1,000)
			600 (A)
Net monetary asset c/f	1,000	0.7	700 (B)
Remeasurement gain (B – A)			100

Exercise 8.2
Answer is (c).

	FC000	Rate	$000
Opening net assets	2,000	0.5	1,000
Net profit for year	800	0.6	480
			$1,480 (A)
Net assets c/f	2,800	0.7	$1,960 (B)
Translation gain (B – A)			480

Exercise 8.3
Answer is (c).

	FC
Net monetary liabilities @ 1.1.20x1	(100,000)
Sales	500,000
Unearned revenue	60,000
Expenses	(350,000)
Issue of new shares	1,000,000
Net monetary assets @ 31.12.20x1	1,110,000

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Exercise 8.4
Answer is (b).

	FC	Rate	$
Net monetary liabilities, 1.1.20x1	(100,000)	0.6	(60,000)
Sales less expenses	150,000	0.55	82,500
Unearned revenue received	60,000	0.48	28,800
Issue of new shares	1,000,000	0.5	500,000
			551,300 (A)
Net assets @ 31.12.20x1	1,110,000	0.5	555,000 (B)
Re-measurement gain (B – A)			3,700

Exercise 8.5

Fair value of net assets:		
Fair value		FC1,430,000
Less deferred tax on fair value differential		(10,000)
Fair value of adjusted net assets		FC1,420,000
30%	=	FC 426,000
Cost of investment in FC (396,000/0.88)	=	FC 450,000
Goodwill in FC	=	FC 24,000
Goodwill in $	=	$21,120

(1) Foreign currency translation reserves attributable to investment in Sanchoz at 31 December 20x2 is -$18,450 (debit balance) obtained as follows:

Translation difference on fair value of net assets

	FC	Exch. Rate	$
Net assets at 31.12.20x1	1,420,000	0.88	1,249,600
Net profit for 20x2	150,000	0.85	127,500
			1,377,100
Net assets at 31.12.20x2	1,570,000	0.84	1,318,800
Translation loss			(58,300)
30% share			(17,490)

Translation difference on goodwill

	FC	Exch. Rate	$
Goodwill at 31.12.20x1	24,000	0.88	21,210
Goodwill at 31.12.20x2	24,000	0.84	20,160
Translation loss on goodwill			(960)

FCTR at 31 December 20x2 = ($17,490) + ($960) = ($18,450)

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(2) Investment in Sanchoz in the consolidated balance sheet is $415,800, derived as follows:

Cost of investment	$396,000
Share of net profit for 20x2	38,250
Share of translation loss	(18,450)
	$415,800

PROBLEMS

Problem 8.1
Note: Year-end is 31 December 20x7.

(1) Transaction gains/losses by each company in the group

Subsidiary 1's transaction exchange gains (losses)

Item	Currency	Amount	Transacted Rate	Closing rate	Exchange gain(loss) in Rmb	Remarks
AR	USD	300,000	8.00	7.67	(99,000)	USD depreciated
AR	Euro	200,000	10.00	9.80	(40,000)	Euro depreciated
AP	AUD	500,000	6.05	5.85	100,000	AUD depreciated
AP	USD	150,000	7.8	7.67	19,500	USD depreciated
AP	USD	130,000	7.9	7.67	29,900	USD depreciated
Total transaction gain(loss)					10,400	

Subsidiary 2's transaction exchange gains (losses)

Item	Currency	Amount	Transacted rate	Closing rate	Exchange gain(loss) in Rmb	Remarks
AR	USD	80,000	1.30	1.31	800	USD depreciated
AP	Rmb	25,000,000	6.00	5.85	(106,838)	Rmb appreciated
AP	Euro	110,000	1.67	1.68	(1,100)	Euro appreciated
					(107,138)	

*AP @ 31 Dec 20x5 = 25,000,000/5.85 = Rmb4,273,504
AP at transaction date = 25,000,000/6 = Rmb4,166,667
Transaction loss Rmb (106,838) [rounding difference of 1]

PT Corp's transaction exchange gains (losses)

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Item	Currency	Amount	Transacted Rate	Closing rate	Exchange gain(loss) in S$	Remarks
AR	Rmb	18,000,000	0.2	0.202	36,364	Rmb appreciated
AR	AUD	250,000	1.2	1.18	(5,000)	AUD depreciated
AR	Euro	150,000	2.00	1.98	(3,000)	Euro depreciated
AR	USD	280,000	1.6	1.55	(14,000)	USD depreciated
AP	Rmb	8,000,000	0.2041	0.202	16,491	Rmb depreciated
AP	AUD	200,000	1.2	1.18	4,000	AUD depreciated
Total transaction gain(loss)					34,855	

(b) Quantify the foreign currency exposure of each company and the group. Note the exposure is to specific currencies.

Currency exposure by Company

Subsidiary 1

USD exposure:	AR	=	US$300,000
	AP	=	(US$280,000)
Net USD exposure			US$ 20,000
Euro exposure	AR	=	Euro200,000
AUD exposure	AP	=	(A$500,000)

Subsidiary 2

USD exposure	AR	=	US$80,000
Euro exposure	AP	=	(Euro110,000)
Rmb exposure	AP	=	(Rmb25,000,000)

PT Corporation

Rmb exposure	AR	=	18,000,000
	AP	=	8,000,000
Net Rmb exposure	(AR)	=	Rmb10,000,000
AUD exposure(Net AR)		=	A$50,000
USD exposure (AR)		=	US$280,000
Euro exposure (AR)		=	Euro 150,000

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Group exposure

	US$	Euro	A$	Rmb
Subsidiary 1	20,000	200,000	(500,000)	0
Subsidiary 2	80,000	(110,000)	0	(25,000,000)
PT Corporation	280,000	150,000	50,000	10,000,000
Total	380,000	240,000	(450,000)	(15,000,000)

Problem 8.2

(1) Income Statement

	Scenario 1: Functional currency is the FC (closing rate method)			**Scenario 2:** Functional currency is the $ (Remeasurement)	
	FC'000	Exch. rate	$'000	Exch. rate	$'000
Sales	87,500	0.28	24,500	0.28	24,500
Cost of sales					
Opening stocks	3,000	0.28	840	0.33	990
Purchases	70,000	0.28	19,600	0.28	19,600
Closing stocks	(5,000)	0.28	(1,400)	0.26	(1,300)
Cost of goods sold	(68,000)		(19,040)		(19,290)
Gross profit	19,500		5,460		5,210
Operating expenses:					
Depreciation	(2,100)	0.28	(588)	0.33	(693)
Other operating expenses	(12,000)	0.28	(3,360)	0.28	(3,360)
Remeasurement difference			Note (a)		243
	(14,100)	0.28	(3,948)		(3,810)
Net profit before tax	5,400	0.28	1,512		1,400
Taxation	(1,600)	0.28	(448)	0.28	(448)
Profit after tax	3,800	0.28	1,064		952

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Note (a): Remeasurement gain

	FC	Exch. Rate	$
Opening monetary liabilities	(4,500)	0.33	(1,485)
Sales	87,500	0.28	24,500
Purchases	(70,000)	0.28	(19,600)
Operating expenses	(12,000)	0.28	(3,360)
Taxation	(1,600)	0.28	(448)
			(393)
Ending monetary liabilities	(600)	0.25	(150)
Remeasurement gain			243

Balance sheet:

	Scenario 1: Functional currency is the FC (closing rate method)			Scenario 2: Functional currency is the dollar (remeasurement)	
	FC'000	Exch. Rate	$'000	Exch. Rate	$'000
Fixed asset (net)	8,400	0.25	2,100	0.33	2,772
Stocks (at cost)	5,000	0.25	1,250	0.26	1,300
Trade debtors	18,000	0.25	4,500	0.25	4,500
Cash	120	0.25	30	0.25	30
	23,120				5,830
Total assets	31,520	0.25	7,880		8,602
Current liabilities	13,720	0.25	3,430	0.25	3,430
Long-term loan	5,000	0.25	1,250	0.25	1,250
Share capital	9,000	0.33	2,970	0.33	2,970
Retained earnings	3,800	I/S	1,064	I/S	952
FCTR		Note (b)	(834)		
Total liabilities and equity	31,520		7,880		8,602

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Note (b): FCTR

	FC	Exch. Rate	$
Opening net assets	9,000	0.33	2,970
Net profit for the year	3,800	0.28	1,064
			4,034
Ending net assets	12,800	0.25	3,200
Translation loss			(834)

(2) The difference in the translation differences stem from the fact that under remeasurement (Temporal Method), the exposed items are monetary items only while under the Closing Rate Method (CRM), all assets and liabilities whether monetary or non-monetary are exposed to translation risk.

In the case of Four Seas Corporation, there is a translation gain because monetary liabilities exceed monetary asset (net monetary liability position) and the FC depreciated against the S$.

On the other hand, the exposed net assets under the CRM resulted in a translation loss.

(3) Impact on ratios

	Remeasurement (TM)	Closing Rate Method
Gross profit margin %	21.26%	22.28%
Operating profit margin % (excluding translation gain)	4.72%	6.17%
Net profit margin %	3.89%	4.34%
ROE	24.27%	33.25%
ROA	13.89%	13.50%

Impact of exchange rate movements on gross profit margin

Using FIFO inventory accounting results in a lag in recognising price level changes. With the local currency depreciating, cost of goods sold will rise faster than sales Under remeasurement, the opening inventory was translated at historical rate which is higher than the average rate used to translate sales. This resulted in a higher COGS and a lower gross profit margin than the CRM.

Operating profit margin

Operating profit margin is lower under remeasurement. In addition to the reason given above, depreciation is higher because fixed assets remains at historical exchange rates. With a

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depreciating local currency, depreciation expense will be higher under remeasurement than under CRM which uses the average rate.

Net profit margin

Under remeasurement, net profit margin will be affected not only by the effects discussed above but also by exchange rate gains or losses on translation. In this case, translation resulted in an exchange gain which however is not large enough to offset the effects of translating certain items at historical cost.

ROE and ROA

Return on equity is higher under the closing rate method because of the higher net profit and lower equity base as a result of the negative translation reserves. Under remeasurement, the equity base is higher because of the effect of exchange rate changes is positive. On the other hand, ROA is slightly higher under remeasurement than under the closing rate method.

Problem 8.3

Minor Company
Translated Income Statement (closing rate method)
for the year ended 31 December 20x2

	FC	Exch. Rate	$
Sales	450,000	0.415	186,750
Cost of sales	220,000	0.415	91,300
Gross profit	230,000	0.415	95,450
Less:			
Depreciation exp	22,500	0.415	9,338
Amortization exp	6,000	0.415	2,490
Insurance exp	800	0.415	332
Operating exp	138,000	0.415	57,270
Taxation exp	12,700	0.415	5,271
	(180,000)	0.415	(74,700)
Profit after tax	50,000	0.415	20,750
Dividends paid	(10,000)	0.42	(4,200)
Dividends paid	(15,000)	0.405	(6,075)
Net profit for the year	25,000		10,475
Retained earnings, 1 Jan	25,000	0.43	10,750
Retained earnings , 31 Dec	50,000		21,225

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Minor Company
Translated Balance Sheet (closing rate method)
as at 31 December 20x2

	FC	Exch. Rate	$
Fixed assets (net)	207,500	0.38	78,850
Patent	54,000	0.38	20,520
Inventories	95,000	0.38	36,100
Prepaid insurance	5,600	0.38	2,128
Accounts receivable	83,900	0.38	31,882
Cash	25,000	0.38	9,500
	471,000	0.38	178,980
Accounts payable	20,000	0.38	7,600
Tax payable	30,000	0.38	11,400
Note payable	20,000	0.38	7,600
Other payables	1,000	0.38	380
Long-term loan	120,000	0.38	45,600
Share capital	230,000	0.43	98,900
Retained earnings	50,000		21,225
FCTR		Note a	(13,725)
Total liabilities and equity	471,000		178,980

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Note (a)

	FC	Exch. Rate	$
Net assets at 1/1/20x2	255,000 *	0.430	109,650
Net profit after tax	50,000	0.415	20,750
Dividends paid	(10,000)	0.420	(4,200)
Dividends paid	(15,000)	0.405	(6,075)
			120,125
Net assets at 31/12/20x2	280,000	0.380	106,400
Translation gain (loss)			(13,725)

*Share capital (FC230,000) + Retained earnings (FC25,000)

Minor Company
Translated Income Statement (closing rate method)
for the year ended 31 December 20x3

	FC	Exch. Rate	$
Sales	500,000	0.36	180,000
Cost of sales	(230,000)	0.36	(82,800)
Gross profit	270,000	0.36	97,200
Expenses	(211,500)	0.36	(76,140)
Net profit after tax	58,500	0.36	21,060
Dividends	(10,000)	0.385	(3,850)
	(15,000)	0.35	(5,250)
Profit retained	33,500		11,960
Retained earnings, 1 Jan	50,000		21,225
Retained earnings, 31 Dec	83,500		33,185

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Minor Company
Translated Balance Sheet (closing rate method)
as at 31 December 20x3

	FC	Exch. Rate	$
Fixed assets (net)	277,500	0.34	94,350
Patent	42,000	0.34	14,280
Current assets:			
Inventories	88,000	0.34	29,920
Prepaid insurance	2,400	0.34	816
Accounts receivable	82,700	0.34	28,118
Cash	13,000	0.34	4,420
	186,100		63,274
	505,600	0.34	171,904
Current liabilities:			
Accounts payable	24,000	0.34	8,160
Tax payable	27,000	0.34	9,180
Note payable	20,000	0.34	6,800
Other payables	1,100	0.34	374
	72,100		24,514
Long-term loan	120,000	0.34	40,800
Share capital	230,000	0.43	98,900
Retained earnings	83,500		33,185
FCTR		Note b	(25,495)
	505,600		171,904

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Note (b)

	FC	Exch. Rate	$
Net assets, 1 Jan 20x3	280,000	0.38	106,400
Net profit	58,500	0.36	21,060
Dividends paid	(10,000)	0.385	(3,850)
Dividends paid	(15,000)	0.35	(5,250)
			118,360
Net assets, 31 Dec 20x3	313,500	0.34	106,590
Translation loss for year			(11,770)
FCTR, 1 Jan 20x3			(13,725)
FCTR, 31 Dec 20x3			(25,495)

Problem 8.4

Minor Company
Remeasured Income Statement (temporal method)
For the year ended 31 December 20x2

	FC	Exch. Rate	$
Sales	450,000	0.415	186,750
COGS	(220,000)	Note (a)	(94,980)
Gross profit	230,000		91,770
Depreciation expense	(22,500)	Note (b)	(9,600)
Amortization expense	(6,000)	0.40	(2,400)
Insurance expense	(800)	0.41	(328)
Operating expense	(138,000)	0.415	(57,270)
Remeasurement gain		Note (c)	1,272
Taxation expense	(12,700)	0.415	(5,271)
Net profit after tax	50,000		18,173
Dividends	(25,000)	Note d	(10,275)

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Retained earnings, 1 Jan	25,000	0.43	10,750
Retained earnings, 31 Dec	75,000		18,648

Note (a): Cost of goods sold

	FC	Exch. Rate	$
Opening inventory	87,000	0.43	37,410
Purchases	228,000	0.415	94,620
	315,000		132,030
Closing inventory	(95,000)	0.39	(37,050)
Cost of goods sold	220,000		94,980

Note (b): Depreciation expense

	Depreciation Exp. (FC)	Exch. Rate	$
Asset acquired on 1.12.20x1	20,000	0.43	8,600
Asset acquired on 10.7.20x2	2,500	0.40	1,000
	22,500		9,600

Note: All assets existing on the date of acquisition of Minor Company are remeasured at the exchange rate on the date of acquisition of Minor Company, that is, 31 December 20x1.

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Note (c): Remeasurement gain

Proof of translation difference:	FC	Exch rate	$
Opening net monetary items	(12,000)*	0.43	(5,160)
Sales	450,000	0.415	186,750
Purchases – inventory	(228,000)	0.415	(94,620)
Purchases – FA	(50,000)	0.40	(20,000)
Purchases – prepaid insurance	(6,400)	0.41	(2,624)
Purchases – patent	(60,000)	0.40	(24,000)
Operating expenses	(138,000)	0.415	(57,270)
Taxation	(12,700)	0.415	(5,271)
Dividend	(10,000)	0.42	(4,200)
	(15,000)	0.405	(6,075)
			(32,470)
Ending net monetary liability	(82,100)	0.38	(31,198)
Re-measurement gain			1,272

*(255,000 – 180,000(fixed assets) – 87,000 (inventory))

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Minor Company

Remeasured Balance sheet (temporal method)

As at 31 December 20x2

	FC	Exch. Rate	$
Fixed assets (net)	207,500	Note (d)	87,800
Patent	54,000	0.4	21,600
Inventories	95,000	0.39	37,050
Prepaid insurance	5,600	0.41	2,296
Accounts receivable	83,900	0.38	31,882
Cash	25,000	0.38	9,500
	471,000		190,128
Accounts payable	20,000	0.38	7,600
Tax payable	30,000	0.38	11,400
Note payable	20,000	0.38	7,600
Other payables	1,000	0.38	380
Long-term loan	120,000	0.38	45,600
Share capital	230,000	0.43	98,900
Retained earnings	50,000		18,648
	471,000		190,128

Note (d): Fixed assets

	Cost (FC)	Net book value(FC)	Exch. Rate	$
Asset acquired on 1.1.20x1	200,000	160,000	0.43	68,800
Asset acquired on 10.7.20x2	50,000	47,500	0.40	19,000
		207,500		87,800

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Minor Company
Remeasured Income Statement (temporal method)
For the year ended 31 December 20x3

	FC	Exch. Rate	$
Sales	500,000	0.36	180,000
Cost of sales	230,000	Note (a)	86,530
Gross profit	270,000		93,470
Less			
Depreciation expense	30,000	Note (b)	(12,450)
Amortization expense	12,000	0.40	(4,800)
Insurance expense	3,200	0.41	(1,312)
Operating expense	152,300	0.36	(54,828)
Taxation expense	14,000	0.36	(5,040)
Remeasurement gain		Note (c)	4,670
	(211,500)		(73,760)
Profit after tax	58,500		19,710
Dividends paid	(10,000)	0.385	(3,850)
Dividends paid	(15,000)	0.35	(5,250)
Retained earnings, 1 Jan	50,000		18,648
Retained earnings, 31 Dec	83,500		29,258

Note (a): Cost of goods sold

	FC	Exch. rate	$
Opening inventory	95,000	0.39	37,050
Purchases	223,000	0.36	80,280
	318,000		117,330
Closing inventory	(88,000)	0.35	(30,800)
	230,000		86,530

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Note (b): Depreciation expense

	Depreciation Exp. (FC)	Exch. rate	$
Asset acquired on 1.1.20x1	20,000	0.43	8,600
Asset acquired on 10.7.20x2	5,000	0.40	2,000
Asset acquired on 1.7.20x3	5,000	0.37	1,850
	30,000		12,450

Note (c): Remeasurement gain

Proof of translation difference	FC	Exch. Rate	$
Opening net monetary liabilities	(82,100)	0.38	(31,198)
Sales	500,000	0.36	180,000
Purchases – inventory	(223,000)	0.36	(80,280)
Purchases – FA	(100,000)	0.37	(37,000)
Operating expenses	(152,300)	0.36	(54,828)
Taxation	(14,000)	0.36	(5,040)
Dividend	(10,000)	0.385	(3,850)
	(15,000)	0.35	(5,250)
			(37,446)
Ending net monetary liabilities	(96,400)	0.34	(32,776)
Remeasurement gain			4,670

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Minor Company
Remeasured Balance sheet (temporal method)
As at 31 December 20x3

	FC	Exch. Rate	$
Fixed assets (net)	277,500	Note (d)	112,350
Patent	42,000	0.4	16,800
Inventories	88,000	0.35	30,800
Prepaid insurance	2,400	0.41	984
Accounts receivable	82,700	0.34	28,118
Cash	13,000	0.34	4,420
Total assets	505,600		193,472
Accounts payable	24,000	0.34	8,160
Tax payable	27,000	0.34	9,180
Note payable	20,000	0.34	6,800
Other payables	1,100	0.34	374
Long-term loan	120,000	0.34	40,800
Share capital	230,000	0.43	98,900
Retained earnings	83,500		29,258
Liabilities + Equity	505,600		193,472

Note (d): Fixed assets

	Cost (FC)	NBV (FC)	Exch. Rate	$
Asset acquired on 1.1.20x1	200,000	140,000	0.43	60,200
Asset acquired on 10.7.20x2	50,000	42,500	0.40	17,000
Asset acquired on 1.7.20x3	100,000	95,000	0.37	35,150
	350,000	277,500		112,350

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A note on accounting treatment of a change in functional currency.

When an entity changes its functional currency, IAS 21 requires all items to be translated using the exchange rate at the date of the change. The example on page 347 in Chapter 7 illustrates this requirement in a highly simplistic way. It should also be pointed out that the example applies to a change in functional currency by a stand-alone entity.

In the case of a group entity, the change in functional currency of a foreign subsidiary is more complex. When a foreign subsidiary is incorporated or acquired, the share capital and pre-acquisition retained earnings, if applicable, are translated at the exchange rate at the date of incorporation or acquisition. This would ensure a 'clean' and complete elimination during the consolidation process. When the share capital and pre-acquisition retained earnings is translated using a rate other than the original exchange rate at the date of incorporation or acquisition, it would have the effect of merging the translation differences in prior period into the translated share capital and retained earnings. In other words, prior periods' cumulative translation differences would disappear. This means that when recording consolidation journal entries to eliminate share capital and pre-acquisition retained earnings against cost of investment, there will not be a clean and complete elimination. To balance the journal entry, a debit or credit entry for the accumulated translation differences has to be included as shown in the following example.

Assume that on 1 January 20x1, a parent company whose functional currency is the dollar incorporated a foreign subsidiary whose functional currency is the FC. The initial paid up share capital is FC1,000 and the exchange rate is 1FC to $1. The cost of investment is therefore $1,000. At 31 December 20x1, the exchange rate is 1FC to $1.10. There is translation gain of $100 which is carried forward to the next year. When the foreign subsidiary's financial statements are consolidated with those of the parent, the following consolidation journal entry is recorded:

Dr	Share capital	1,000	
Cr	Cost of investment		1,000

Assume that on 1 January 20x2, the functional currency of the foreign subsidiary is changed to the dollar. Under IAS 21, the share capital is translated at FC1:$1.10 which is equal to $1,100. This is more than the original cost of investment of $1,000 and the consolidation journal entry will not balance. To 'force' the entry to balance, it is necessary to 'restore' the prior period's translation difference as follows:

Dr	Share capital	1,100	
Cr	Cost of investment		1,000
Cr	Translation reserves		100

The translation reserves will flow through to the consolidated balance sheet. This is consistent with the requirement of IAS 21:37. An alternative approach, which is the one adopted for **Problems 7.5 and 7.6**, is to translate the share capital and pre-acquisition retained earnings, where applicable, at the original exchange rates. The cumulative translation differences will be unaffected and appears in the consolidated balance sheet.

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Problem 8.5

The following points should be noted:

(1) The method of translation is changed from the closing rate method (in 20x2) to the temporal method (20x3).
(2) The change in functional currency is to be accounted for prospectively from the date of the change (IAS 21:37)
(3) At the date of the change, the translated amounts for non-monetary items carried at cost are treated as their historical costs.
(4) The cumulative translation gains (losses) are retained in equity until the foreign operation is disposed of.

Minor Company
Translated Balance Sheet (closing rate method)
as at 31 December 20x2

	FC	Exch. Rate	$
Fixed assets (net)	207,500	0.38	78,850
Patent	54,000	0.38	20,520
Inventories	95,000	0.38	36,100
Prepaid insurance	5,600	0.38	2,128
Accounts receivable	83,900	0.38	31,882
Cash	25,000	0.38	9,500
	471,000	0.38	178,980
Accounts payable	20,000	0.38	7,600
Tax payable	30,000	0.38	11,400
Note payable	20,000	0.38	7,600
Other payables	1,000	0.38	380
Long-term loan	120,000	0.38	45,600
Share capital	230,000	0.43	98,900
Retained earnings	50,000		21,225
FCTR		Note a	(13,725)
Total liabilities and equity	471,000		178,980

Note (a)

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	FC	Exch. Rate	$
Net assets at 1/1/20x2	255,000 *	0.430	109,650
Net profit after tax	50,000	0.415	20,750
Dividends paid	(10,000)	0.420	(4,200)
Dividends paid	(15,000)	0.405	(6,075)
			120,125
Net assets at 31/12/20x2	280,000	0.380	106,400
Translation gain (loss)			(13,725)

*Share capital (FC230,000) + Retained earnings (FC25,000)

Minor Company
Remeasured Income Statement (temporal method)
For the year ended 31 December 20x3

	FC	Exch. Rate	$
Sales	500,000	0.36	180,000
Cost of sales	230,000	(a)	85,580
Gross profit	270,000		94,420
less			
Depreciation expense	30,000	(b)	(11,350)
Amortization expense	12,000	0.38	(4,560)
Insurance expense	3,200	0.38	(1,216)
Operating expense	152,300	0.36	(54,828)
Taxation expense	14,000	0.36	(5,040)
Remeasurement gain		Note (c)	4,670
	(211,500)		(72,324)
Profit after tax	58,500		22,096
Dividends	(10,000)	0.385	(3,850)
	(15,000)	0.35	(5,250)
Retained earnings, 1 Jan	50,000		21,225
Retained earnings, 31 Dec	83,500		34,221

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Note (a): Cost of goods sold

	FC	Exch. Rate	$
Opening inventory	95,000	0.38	36,100
Purchases	223,000	0.36	80,280
	318,000		116,380
Closing inventory	(88,000)	0.35	(30,800)
	230,000		85,580

Note (b): Depreciation expense

	Depreciation Exp. (FC)	Exch. rate	$
Asset acquired on 1.1.20x1	20,000	0.38	7,600
Asset acquired on 10.7.20x2	5,000	0.38	1,900
Asset acquired on 1.7.20x3	5,000	0.37	1,850
	30,000		11,350

Note (c) : Remeasurement gain

Proof of translation difference

	FC	Exch. Rate	$
Opening net monetary items	(82,100)	0.38	(31,198)
Sales	500,000	0.36	180,000
Purchases - inventory	(223,000)	0.36	(80,280)
Purchases – FA	(100,000)	0.37	(37,000)
Operating expenses	(152,300)	0.36	(54,828)
Taxation	(14,000)	0.36	(5,040)
Dividend paid	(10,000)	0.385	(3,850)
Dividend paid	(15,000)	0.35	(5,250)
			(37,446)
	(96,400)	0.34	(32,776)
Remeasurement gain on monetary items			4,670

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Minor Company

Remeasured Balance sheet (temporal method)
As at 31 December 20x3

	FC	Exch. Rate	$
Fixed assets (net)	277,500	Note (d)	104,500
Patent	42,000	0.38	15,960
Inventories	88,000	0.35	30,800
Prepaid insurance	2,400	0.38	912
Accounts receivable	82,700	0.34	28,118
Cash	13,000	0.34	4,420
Total assets	505,600		184,710
Accounts payable	24,000	0.34	8,160
Tax payable	27,000	0.34	9,180
Note payable	20,000	0.34	6,800
Other payables	1,100	0.34	374
Long-term loan	120,000	0.34	40,800
Share capital	230,000	0.43	98,900
Retained earnings	83,500		34,221
FCTR			(13,725)
Liabilities + Equity	505,600		184,710

Note (d): Fixed assets

	Cost (FC)	NBV (FC)	Exch. Rate	$
Asset acquired on 1.1.20x1	200,000	140,000	0.38	53,200
Asset acquired on 10.7.20x2	50,000	42,500	0.38	16,150
Asset acquired on 1.7.20x3	100,000	95,000	0.37	35,150
	350,000	277,500		104,500

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Problem 8.6

The following points should be noted:

(1) The method of translation is changed from the temporal method (in 20x2) to the closing rate method (20x3).
(2) The change in functional currency is to be accounted for prospectively from the date of the change (IAS 21:37)

Minor Company

Remeasured Income Statement (temporal method)

For the year ended 31 December 20x2

	FC	Exch. Rate	$
Sales	450,000	0.415	186,750
Cost of sales	(220,000)	Note (a)	(94,980)
Gross profit	230,000		91,770
less			
Depreciation expense	(22,500)	Note (b)	(9,600)
Amortization expense	(6,000)	0.40	(2,400)
Insurance expense	(800)	0.41	(328)
Operating expense	(138,000)	0.415	(57,270)
Remeasurement gain		Note (c)	1,272
Taxation exp	(12,700)	0.415	(5,271)
Net profit after tax	50,000		18,173
Dividends	(25,000)	Note (d)	(10,275)
Retained earnings, 1 Jan	25,000	0.43	10,750
Retained earnings, 31 Dec	75,000		18,648

Note (a): Cost of goods sold

	FC	Exch. Rate	$
Opening inventory	87,000	0.43	37,410
Purchases	228,000	0.415	94,620
	315,000		132,030
Closing inventory	(95,000)	0.39	(37,050)
Cost of goods sold	220,000		94,980

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Note (b): Depreciation expense

	Depreciation Exp. (FC)	Exch. Rate	$
Asset acquired on 1.12.20x1	20,000	0.43	8,600
Asset acquired on 10.7.20x2	2,500	0.40	1,000
	22,500		9,600

Note: All assets existing on the date of acquisition of Minor Company are remeasured at the exchange rate on the date of acquisition of Minor Company, that is, 31 December 20x1.

Note (c): Remeasurement gain

Proof of translation difference	FC	Exch rate	$
Opening net monetary liabilities	(12,000)	0.43	(5,160)
Sales	450,000	0.415	186,750
Purchases – inventory	(228,000)	0.415	(94,620)
Purchases – FA	(50,000)	0.40	(20,000)
Purchases – prepaid insurance	(6,400)	0.41	(2,624)
Purchases – patent	(60,000)	0.40	(24,000)
Operating expenses	(138,000)	0.415	(57,270)
Taxation	(12,700)	0.415	(5,271)
Dividend	(10,000)	0.42	(4,200)
	(15,000)	0.405	(6,075)
			(32,470)
Ending net monetary liabilities	(82,100)	0.38	(31,198)
Remeasurement gain			1,272

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Minor Company

Remeasured Balance sheet (temporal method)

As at 31 December 20x2

	FC	Exch. Rate	$
Fixed assets (net)	207,500	Note (d)	87,800
Patent	54,000	0.4	21,600
Inventories	95,000	0.39	37,050
Prepaid insurance	5,600	0.41	2,296
Accounts receivable	83,900	0.38	31,882
Cash	25,000	0.38	9,500
	471,000		190,128
Accounts payable	20,000	0.38	7,600
Tax payable	30,000	0.38	11,400
Note payable	20,000	0.38	7,600
Other payables	1,000	0.38	380
Long-term loan	120,000	0.38	45,600
Share capital	230,000	0.43	98,900
Retained earnings	50,000		18,648
	471,000		190,128

Note (d): Fixed assets

	Cost (FC)	Net book value(FC)	Exch. Rate	$
Asset acquired on 1.1.20x1	200,000	160,000	0.43	68,800
Asset acquired on 10.7.20x2	50,000	47,500	0.40	19,000
		207,500		87,800

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Minor Company
Translated Income Statement (closing rate method)
for the year ended 31 December 20x3

	FC	Exch. Rate	$
Sales	500,000	0.36	180,000
Cost of sales	(230,000)	0.36	(82,800)
Gross profit	270,000	0.36	97,200
Expenses	(211,500)	0.36	(76,140)
Net profit after tax	58,500	0.36	21,060
Dividends paid	(10,000)	0.385	(3,850)
Dividends paid	(15,000)	0.35	(5,250)
Profit retained	33,500		11,960
Retained earnings, 1 Jan	50,000		18,648
Retained earnings, 31 Dec	83,500		30,608

Minor Company
Translated Balance Sheet (closing rate method)
as at 31 December 20x3

	FC	Exch. Rate	$
Fixed assets (net)	277,500	0.34	94,350
Patent	42,000	0.34	14,280
Inventories	88,000	0.34	29,920
Prepaid insurance	2,400	0.34	816
Accounts receivable	82,700	0.34	28,118
Cash	13,000	0.34	4,420
	186,100		63,274
	505,600	0.34	171,904

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Accounts payable	24,000	0.34	8,160
Tax payable	27,000	0.34	9,180
Note payable	20,000	0.34	6,800
Other payables	1,100	0.34	374
	72,100		24,514
Long-term loan	120,000	0.34	40,800
Share capital	230,000	0.43	98,900
Retained earnings	83,500		30,608
FCTR		Note b	(22,918)
	505,600		171,904

Translation difference due to change of functional currency on 1 January 20x3:

Net assets b/f, in dollars	$117,548
Net assets b/f , translated at exchange rate on date of change	$106,400 (FC280,000 x 0.38)
Translation loss	($11,148)

	FC	Exch. Rate	$
Net assets at 1/1/20x3	280,000	0.38	106,400
Profit after tax for 20x3	58,500	0.36	21,060
Dividends paid	(10,000)	0.385	(3,850)
Dividends paid	(15,000)	0.35	(5,250)
			118,360
Net assets at 31/12/20x3	313,500	0.34	106,590
Translation loss for 20x3			(11,770)
Add: Translation loss on date of change			(11,148)
Foreign Currency Translation Reserves c/f			(22,918)

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Problem 8.7

(1) Translation of the financial statements of AMCO Inc for the year 20x3 and 20x4. (Use Closing Rate method since AMCO's functional currency is the US dollar)

AMCO Inc.
Income Statement
For the year ended 31 December

	20x4			**20x3**		
	US$000	Exch. Rate	S$000	US$000	Exch. rate	S$000
Sales	48,000	1.72	82,560	42,000	1.75	73,500
Cost of sales	(36,000)	1.72	(61,920)	(29,000)	1.75	(50,750)
Gross profit	12,000	1.72	20,640	13,000	1.75	22,750
Operating expense	(7,000)	1.72	(12,040)	(6,000)	1.75	(10,500)
Profit before tax	5,000	1.72	8,600	7,000	1.75	12,250
Taxation	(2,000)	1.72	(3,440)	(3,000)	1.75	(5,250)
Net profit	3,000	1.72	5,160	4,000	1.75	7,000
Retained earnings, 1 Jan	35,000	Note (a)	62,490	31,000	Note(a)	55,490
Dividends paid	(13,000)	1.68	(21,840)			
Retained earnings, 31 Dec	25,000		45,810	35,000		62,490

Note (a): Retained earnings (Figures in $000)

At 1 January 20x2	0
Net profit for 20x2 (US$31,000 x $1.79)	= S$55,490
Retained earnings at 1 January 20x3	= S$55,490
Net profit for 20x3 (US$4,000 x 1.75)	= S$7,000
Retained earnings at 1 January 20x4	= S$62,490

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AMCO Inc

Balance sheets
As at 31 December

	20x4			20x3		
	US$000	Exch. rate	S$000	US$000	Exch. rate	S$000
Fixed assets	200,000	1.68	336,000	185,000	1.73	320,050
Current assets	200,000	1.68	336,000	250,000	1.73	432,500
Current liabilities	(150,000)	1.68	(252,000)	(170,000)	1.73	(294,100)
Long-term liabilities	(125,000)	1.68	(210,000)	(130,000)	1.73	(224,900)
Net assets	125,000		210,000	135,000		233,550
Represented by:						
Share capital	100,000	1.8	180,000	100,000	1.8	180,000
Retained earnings	25,000		45,810	35,000		62,490
FCTR		Note(b)	(15,810)		Note (b)	(8,940)
Shareholders' equity	125,000		210,000	135,000		233,550

Note (b): Foreign currency translation reserves

20x2	US$000	Exch. rate	S$000
Net assets, 1 Jan	100,000	1.8	180,000
Net profit for year	31,000	1.79	55,490
			235,490
Net assets, 31 Dec	131,000	1.78	233,180
Translation gain(loss)			(2,310)

20x3	US$000	Exch rate	S$000
Net assets, 1 Jan	131,000	1.78	233,180
Net profit for the year	4,000	1.75	7,000
			240,180
Net assets, 31 Dec	135,000	1.73	233,550
Translation gain(loss)			(6,630)

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20x4	US$000	Exch rate	S$000
Net assets, 1 Jan	135,000	1.73	233,550
Net profit for the year	3,000	1.72	5,160
Dividend paid	(13,000)	1.68	(21,840)
			216,870
Net assets, 31 Dec	125,000	1.68	210,000
Translation gain(loss)			(6,870)

Cumulative Foreign Currency Translation Reserves (FCTR)

	S$000
20x2	(2,310)
20x3	(6,630)
20x4	(6,870)
Total	15,810

2. Foreign Currency Reserves in Asian Macro's consolidated balance sheet at 31.12.2004:

	S$000
Translation reserves of AMCO (15,810 x 0.8)	(12,648)
Share of translation reserves of OZCO (Note c)	720
	(11,928)

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Note (c)

Translation reserves – OZCO

	A$000	Exch. Rate	S$000
Net assets at 31.12.20x2	20,000	1.13	22,600
Net profit for 20x3	2,000	1.15	2,300
			24,900
Net asset at 31.12.20x3	22,000	1.16	25,520
Translation gain (loss) for 20x3			620
Net asset at 31.12.20x3	22,000	1.16	25,520
Net profit for 20x4	2,000	1.18	2,360
Revaluation surplus	2,000	1.20	2,400
Dividends paid	(1,000)	1.21	(1,210)
			29,070
Net asset at 31.12.20x4	25,000	1.21	30,250
Translation gain (loss) for 20x4			1,180
Cumulative translation differences (620 + 1,180)			1,800
40 % share of FCTR			720

3. Investment in OZCO Corporation in Asian Macro's consolidated balance sheet at 31 December 20x4.

Cost of investment	A$8,000,000 @ 1.13 = $9,040,000
Share of net profit:	
20x3 40% x A$2,000,000 x 1.15	920,000
20x4 40% x A$2,000,000 x 1.18	944,000
Share of revaluation surplus 40% x A$2,000,000 x 1.20	960,000
Share of translation reserves (see Note c)	720,000
Less dividend 40% x A$1,000,000 x 1.21	(484,000)
Investment in OZCO at 31.12.20x4	$12,100,000

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Problem 8.8

(1) As the functional currency of Evergreen Ltd is the FC, the closing rate method is used to translate the financial statements.

Evergreen Ltd
Profit & Loss Account
For the year ended 31 October 20x3

	FC	Rate	S$
Sales	7,800,000	0.75	5,850,000
Cost of sales	4,835,000	0.75	3,626,250
Gross profit	2,965,000	0.75	2,223,750
Operating expenses	(2,300,000)	0.75	(1,725,000)
Depreciation	(375,000)	0.75	(281,250)
Amortisation expense	(50,000)	0.75	(37,500)
Exchange gain	50,000	0.80 *	40,000
Profit before tax	290,000		220,000
Taxation	(87,000)	0.75	(65,250)
Profit after tax	203,000		154,750
Retained earnings, 1 Jan	311,000	Note (a)	242,030
Dividends	(50,000)	0.77	(38,500)
Retained earnings, 31 Dec	464,000		358,280

Note (a): Retained earnings, 1 January is obtained by reconstructing the opening balance sheet.

	FC	Exch. Rate	S$
Net assets, 1 Jan	2,811,000	0.73	2,052,030
Share capital	2,500,000	Note (b)	1,810,000
Retained earnings	311,000	Balancing figure	242,030
	2,811,000		2,052,030

Note (b): Share capital

	FC	Exch. Rate	$
First issue	1,500,000	0.7	1,050,000
Second issue	1,000,000	0.76	760,000
	2,500,000		1,810,000

Evergreen Ltd
Balance sheet

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At 31 October 20x3

	FC	Exch. Rate	$
Fixed assets	3,920,000	0.80	3,136,000
Patent	200,000	0.80	160,000
Current assets:			
Inventory	600,000	0.80	480,000
Deposits	100,000	0.80	80,000
Receivables	979,000	0.80	783,200
Cash	80,000	0.80	64,000
			4,703,200
Current liabilities			
Accounts payable	(878,000)	0.80	(702,400)
Tax payable	(87,000)	0.80	(69,600)
Loan payable	(1,950,000)	0.80	(1,560,000)
	2,964,000		2,371,200
Share capital	2,500,000	Note (a)	1,810,000
Retained earnings	464,000		358,280
Translation reserves		Note (c)	202,920
	2,964,000		2,371,200

Note: You can assume that interest on intercompany loan is paid on 31 December.

Note (c): Translation gains (losses)

	FC	Exch. Rate	$
Net assets, 1 Jan	2,811,000*	0.73	2,052,030
Net profit	203,000		154,750
Dividend paid	(50,000)	0.77	(38,500)
			2,168,280
Net assets, 31 Dec	2,964,000	0.80	2,371,200
Translation gain(loss)			202,920

*Share capital (2,500,000) + retained earnings (311,000)

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(2) Consolidation Journal Entries

CJE 1	DR	Share capital	1,810,000	
	CR	Investment in Evergreen		1,810,000
CJE 2	DR	Dividend income	38,500	
	CR	Dividend declared		38,500
CJE 3	DR	Loan receivable	1,560,000	
	CR	Loan payable		1,560,000
CJE 4	DR	Interest income (Blue Sky)	78,000*	
	CR	Interest expense (Evergreen)		78,000

*It is assumed that the interest income has been included under operating expenses (as a credit item) in Blue Sky's income statement.

(3) Consolidated financial statements

	Blue Sky	Evergreen	Consolidation Adjustments		Consolidated Income Statement
			Dr	Cr	
	$	$	$	$	
Sales	8,750,000	5,850,000			14,600,000
Cost of sales	(6,800,000)	(3,626,250)			(10,426,250)
Gross profit	1,950,000	2,223,750			4,173,750
Operating expenses	(1,410,000)	(1,725,000)	78,000	78,000	(3,135,000)
Depreciation		(281,250)			(281,250)
Amortisation expense		(37,500)			(37,500)
Exchange gain		40,000			40,000
Dividend income	38,500		38,500		
Profit before tax	578,500	220,000			760,000
Taxation	(143,500)	(65,250)			(208,750)
Profit after tax	435,000	154,750			551,250
Retained earnings, 1 Jan	2,052,000	242,030			2,294,030
Dividend		(38,500)		38,500	
Retained earnings, 31 Dec	2,487,000	358,280	116,500	116,500	2,845,280

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	Blue Sky	Evergreen	Adjustments		Consolidated Consolidated
	S$	S$	Dr	Cr	Balance sheet
Fixed assets	8,400,000	3,136,000			11,536,000
Patent		160,000			160,000
Investment in Evergreen	1,810,000			1,810,000	
Loan receivable	1,560,000			1,560,000	
Current assets:		-			
Inventory	500,000	480,000			980,000
Deposits		80,000			80,000
Receivables	1,800,000	783,200			2,583,200
Cash	50,000	64,000			114,000
	14,120,000	4,703,200			15,453,200
Current liabilities					
Accounts payable	1,495,000	702,400			2,197,400
Tax payable	138,000	69,600			207,600
Loan payable		1,560,000	1,560,000		
Share capital	10,000,000	1,810,000	1,810,000		10,000,000
Retained earnings	2,487,000	358,280	116,500	116,500	2,845,280
Translation reserves		202,920			202,920
	14,120,000	4,703,200	3,486,500	3,486,500	15,453,200

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Problem 8.9

(1) As the functional currency of Evergreen is the dollar, remeasurement (temporal method) is used.

Evergreen Ltd
Income Statement
For the year ended 31 October 20x3

	FC	Exch. Rate	S$
Sales	7,800,000	0.75	5,850,000
Cost of sales	4,835,000	Note (a)	3,587,250
Gross profit	2,965,000	0.75	2,262,750
Operating expenses	(2,300,000)	0.75	(1,725,000)
Depreciation	(375,000)	Note (b)	(276,150)
Amortization expense	(50,000)	0.76	(38,000)
Exchange gain	50,000	0.80	40,000*
Remeasurement loss		Note (c)	(33,980)
Profit before tax	290,000		229,620
Taxation	(87,000)	0.75	(65,250)
Profit after tax	203,000		164,370
Retained earnings, 1 Jan	311,000	Note (d)	264,030
Dividend	(50,000)	0.77	(38,500)
Retained earnings, 31 Dec	464,000		389,900

*Note: A case may be argued for using average rate for exchange gain

Note (a): Cost of sales

	FC	Exch. Rate	$
Opening inventory	500,000	0.72	360,000
Purchases	4,935,000	0.75	3,701,250
Closing inventory	(600,000)	0.79	(474,000)
COGS	4,835,000		3,587,250

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Note (b): Depreciation expense

Depreciation expense (FC)	Exch rate	Depreciation expense ($)
30,000	0.7	21,000
120,000	0.72	86,400
225,000	0.75	168,750
375,000		276,150

Note (c): Exchange gain (loss)

Under the remeasurement (temporal method) approach, the foreign operation's transactions are treated as though they were the transactions of the parent. In other words, they are deemed as foreign currency transactions of the parent. Consequently, remeasurement gains (or losses) are taken to the income statement in the same way as transaction exchange gains or losses. In practice, the remeasurement and transaction gains (or losses) are combined and reported as a single item – exchange gains (or losses).

	FC	Exchange rate	$
Net monetary liabilities, 1 Jan	(234,000)	Note (d)	(170,820)
Sales	7,800,000	0.75	5,850,000
Purchases	(4,935,000)	0.75	(3,701,250)
Taxation	(87,000)	0.75	(65,250)
Operating expenses	(2,300,000)	0.75	(1,725,000)
Exchange gain	50,000	0.80	40,000*
Dividend	(50,000)	0.77	(38,500)
Purchase of fixed asset	(2,000,000)	0.78	(1,560,000)
			(1,370,820)
Net monetary liabilities, 31 Dec	(1,756,000)	0.80	(1,404,800)
Remeasurement loss			(33,980)
Add: Exchange gain on intercompany loan			40,000
Net exchange gain (loss)			6,020

Note (d): Opening monetary liabilities and retained earnings

The opening net monetary position is not given but can be derived by reconstructing the opening balance sheet as follows:

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	FC	Exch. Rate	$
Fixed assets	2,295,000	Note (e)	1,694,850
Patent	250,000	0.76	190,000
Inventory	500,000	0.72	360,000
Net monetary liabilities	(234,000)	0.73	(170,820)
	2,811,000		2,074,030
Represented by:			
Share capital @ 31/10/20x2	2,500,000		1,810,000
RE - b/f	311,000	Bal. figure	264,030
	2,811,000		2,074,030

The opening net monetary liabilities is obtained as a balancing and translated at the closing rate.

*Conceptual note: As the loan is denominated in dollars, from the group's point of view, there should not be any exchange gain or loss in one's own currency. The exchange gain or loss may be remeasured at nil (zero exchange rate). The net effect is the same as remeasurement results in a gain of $6,020 (and a nil transaction exchange gain).

Note (e): Opening fixed assets

Cost(FC)	Accum dep.	Net book value	Exchange rate	$
300,000	(60,000)	240,000	0.70	168,000
600,000	(120,000)	480,000	0.72	345,600
1,800,000	(225,000)	1,575,000	0.75	1,181,250
2,700,000	(405,000)	2,295,000		1,694,850

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Evergreen Ltd
Balance sheet
As at 31 October 20x3

	FC	Exch rate	$
Fixed assets	3,920,000	Note (f)	2,978,700
Patent	200,000	0.76	152,000
Current assets:			
Inventory	600,000	0.79	474,000
Deposits	100,000	0.80	80,000
Receivables	979,000	0.80	783,200
Cash	80,000	0.80	64,000
	1,759,000		4,531,900
Current liabilities			
Accounts payable	(878,000)	0.80	(702,400)
Tax payable	(87,000)	0.80	(69,600)
Loan payable	(1,950,000)	0.80	(1,560,000)
	2,964,000		2,199,900
Share capital	2,500,000		1,810,000
Retained earnings	464,000		389,900
	2,964,000		2,199,900

Note (f): Fixed assets

Cost(FC)	Accumulated Depreciation	NBV	Exch. Rate	$
300,000	(90,000)	210,000	0.70	147,000
600,000	(240,000)	360,000	0.72	259,200
1,800,000	(450,000)	1,350,000	0.75	1,012,500
2,000,000		2,000,000	0.78	1,560,000
4,700,000	(780,000)	3,920,000		2,978,700

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(2) Evergreen Ltd
Consolidated Income Statement
For the year ended 31 October 20x3

	Blue Sky S$	Evergreen S$	Consolidation Adjustments Dr	Cr	Consolidated Income Statement
Sales	8,750,000	5,850,000			14,600,000
COGS	(6,800,000)	(3,587,250)			(10,387,250)
Gross profit	1,950,000	2,262,750			4,212,750
Operating expenses	(1,410,000)	(1,725,000)	78,000	78,000	(3,135,000)
Depreciation		(276,150)			(276,150)
Amortisation expense		(38,000)			(38,000)
Exchange gain		6,020			6,020*
Dividend income	38,500		38,500		
Profit before tax	578,500	229,620			769,620
Taxation	(143,500)	(65,250)			(208,750)
Profit after tax	435,000	164,370			560,870
Retained earnings, 1 Jan	2,052,000	264,030			2,316,030
Dividends		(38,500)		38,500	
Retained earnings, 31 Dec	2,487,000	389,900	116,500	116,500	2,876,900

*For presentation purposes, remeasurement loss and transaction gain on the intercompany loan have been netted off.

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Evergreen Ltd
Consolidated Balance Sheet
At 31 October 20x3

	Blue Sky	Evergreen	Adjustments		Consolidated
	$	$	Dr	Cr	Balance sheet
Fixed assets	8,400,000	2,978,700			11,378,700
Patent		152,000			152,000
Investment in Evergreen	1,810,000			1,810,000	-
Loan receivable	1,560,000			1,560,000	-
Current assets:					
Inventory	500,000	474,000			974,000
Deposits		80,000			80,000
Receivables	1,800,000	783,200			2,583,200
Cash	50,000	64,000			114,000
	1,850,000	1,401,200			3,751,200
Total assets	14,120,000	4,531,900			15,281,900
Current liabilities					
Accounts payable	1,495,000	702,400			2,197,400
Tax payable	138,000	69,600			207,600
Loan payable		1,560,000	1,560,000		
Share capital	10,000,000	1,810,000	1,810,000		10,000,000
Retained earnings	2,487,000	389,900	116,500	116,500	2,876,900
Total liabilities and equity	14,120,000	4,531,900	3,486,500	3,486,500	15,281,900

Problem 8.10

(a) Fixed assets	Total	P&M	Motor vehicles	SGD
Cost	900,000	800,000	100,000	1,543,000
Less acc dep	(245,000)	(240,000)	(5,000)	(419,150)
Net book value	655,000	560,000	95,000	1,123,850

(b) Depreciation expense	USD	Exch. rate	SGD
Plant and machinery	190,000	1.71	324,900
Motor vehicles	5,000	1.75	8,750
	195,000		333,650

(c) Profit on sale of fixed asset

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	USD	Exch. Rate	SGD
Cost	200,000		
Accumulated depreciation	50,000		
Net book value	150,000	1.71	256,500
Proceeds	180,000	1.78	320,400
Profit on sale	30,000		63,900
(d) Cost of sales			
Opening stock	100,000	1.68	168,000
Purchases	1,500,000	1.75	2,625,000
Closing stock	(200,000)	1.78	(356,000)
Cost of sales			2,437,000
(e) Available-for-sale securities (Asset)			
Balance as at 1 Jan 20x5	880,000	1.70	1,496,000
Fair value adjustment at year-end	120,000	1.79	214,800
Balance per movement			1,710,800
Increase due to exchange gain			79,200
Balance as at 31 Dec 20x5	1,000,000	1.79	1,790,000
(f) Fair value adjustment reserve (equity)			
Balance in fair value reserve (equity) at 1.1.20x5			(52,000)
Translation difference taken to fair value reserve in equity			79,200
Change in fair value of AFS			214,800
Fair value adjustment taken direct to equity			242,000
(g) Trading securities (Asset)			
Balance as at 1 Jan 20x5	0		0
Purchase during 20x5	350,000	1.74	609,000
Fair value adjustment at year-end	50,000	1.79	89,500
Balance per movement			698,500
Translation gain			17,500
Balance as at 31 Dec 20x5	400,000	1.79	716,000
(h) Exchange gain relating to fair value adjustment			17,500
Fair value adjustment			89,500
Fair value adjustment taken to P&L			107,000
(i) Prepaid Insurance			
Prepaid insurance	45,000	1.78	80,100
(j) Insurance expense			
For period 1 Jan to 30 Sept 20x5	37,500	1.73	64,875

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For period 1 Oct to 31 Dec 20x5	15,000	1.78	26,700
	52,500		91,575

Exchange gain arising from movement of net monetary liabilities

	USD	Rate	SGD
Net monetary liabilities at 1 Jan 20x5	(700,000)	1.70	(1,190,000)
Movements during 20x5			
Sales	2,000,000	1.75	3,500,000
Purchases	(1,500,000)	1.75	(2,625,000)
Operating expenses	(260,000)	1.75	(455,000)
Interest expense	(50,000)	1.75	(87,500)
Tax expense	(20,000)	1.75	(35,000)
Purchase of trading security	(350,000)	1.74	(609,000)
Purchase of motor vehicles	(100,000)	1.75	(175,000)
Sale of fixed assets	180,000	1.78	320,400
Prepayment of fire insurance	(60,000)	1.78	(106,800)
Balance per movement	(860,000)		(1,462,900)
Net monetary liabilities at 31 Dec 20x5			
	(860,000)	1.79	(1,539,400)
Exchange loss in income statement relating to movement in monetary items			(76,500)

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Problem 8.11

(1) Translated financial statements

Income Statement for the year ended 31 Dec 20x5

	$	Rate	FC
Profit before tax	1,300,000	1.25	1,625,000
Tax	-260,000	1.25	-325,000
Profit after tax	1,040,000		1,300,000
Dividends declared	-200,000	1.28	-256,000
Profit retained	840,000		1,044,000
Retained earnings, 1 Jan 20x5	1,500,000	Working	1,700,000
Retained earnings, 31 Dec 20x5	2,340,000		2,744,000

Working: Reconstructed Balance sheet as at 1 Jan 20x5

Net assets	2,000,000	1.20	2,400,000
Share capital	500,000	1.10	550,000
Retained earnings	1,500,000	Residual	1,700,000
FCTR		Given	150,000
	2,000,000		2,400,000

Balance sheet as at 31 Dec 20x5	**$**		**FC**
Fixed assets, net book value	2,300,000	1.3	2,990,000
Inventory	500,000	1.3	650,000
Accounts receivable	150,000	1.3	195,000
Cash	100,000	1.3	130,000
	3,050,000		3,965,000
Less accounts payable	-210,000	1.3	-273,000
Net assets	2,840,000	1.30	3,692,000
Share capital	500,000	1.10	550,000
Retained earnings pre-acquisition	1,000,000	1.10	1,100,000
Retained earnings post-acquisition	1,340,000		1,644,000
Foreign Currency Translation Reserve (FCTR)		Note(a)	398,000
	2,840,000		3,692,000

Note (a):

Movement in net assets	$	Exch. Rate	FC
Net assets at 1 Jan 20x5	2,000,000	1.2	2,400,000
Profit for the year	1,040,000	1.25	1,300,000
Dividends declared	-200,000	1.28	-256,000
	2,840,000		3,444,000
Change in FCTR			248,000

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Net assets at 31 Dec 20x5	2,840,000	1.3	3,692,000
FCTR, 1 Jan, from book value of net assets			150,000
Change in FCTR			248,000
FCTR, 31 Dec, from book value of net assets			398,000
Investment in Y Co	1,800,000	1.10	1,980,000

(2) Prepare the consolidation adjusting entries for the year ended 31 Dec 2005 in FC (FC is the presentation currency).

Consolidation adjustments

CJE1: Eliminate investment in Y Co				
Dr Share capital		550,000		
Dr Retained earnings		1,100,000		
Dr Goodwill		352,000		
Dr Intangible asset		220,000		
Cr Deferred tax liability				44,000
Cr Investment in Y Co				1,980,000
Cr Non-controlling interests				198,000
Parent's share of goodwill		$ 306,000 (FC 306,000)		
NCI's share of goodwill		$ 14,000 (FC 336,000)		
CJE2: Cumulative FCTR on goodwill				
Dr Goodwill		64,000		
Cr FCTR				64,000
Goodwill	$	320,000	1.1	352,000
Goodwill at closing rate	$	320,000	1.3	416,000
Change in FCTR due to goodwill				64,000
NCI's share of goodwill		14,000	1.1	15,400
NCI's share of goodwill at closing rate		14,000	1.3	18,200
				2,800
CJE3: Cumulative FCTR on fair value adjustment (intangible asset)				
Dr Intangible asset		40,000		
Cr FCTR				40,000
Intangible asset	$	200,000	1.10	220,000
Intangible asset at closing rate	$	200,000	1.3	260,000
Change in FCTR due to intangible asset				40,000

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CJE4: Cumulative FCTR on deferred tax liability on intangible asset

Dr FCTR		8,000		
Cr Deferred tax liability				8,000
Deferred tax liability	$	40,000	1.10	44,000
Deferred tax liability at closing rate	$	40,000	1.30	52,000

CJE5: Allocation of FCTR to non-controlling interests

Dr FCTR		45,800		
Cr Minority interests				45,800
Non-controlling interests have a share of:				
FCTR from book value of net assets				398,000
FCTR from intangible asset				40,000
FCTR from deferred tax liability on intangible asset				-8,000
				430,000
NCI's share (10%)				43,000
NCI's share of FCTR on goodwill				2,800
				45,800

Workings:

Balance sheet at 31 Dec 20x4

	$	**Rate**	**FC**
Net assets	2,000,000	1.20	2,400,000
Share capital	500,000	1.10	550,000
Retained earnings-pre-acquisition	1,000,000	1.10	1,100,000
Retained earnings -post acquisition	500,000	Residual	600,000
			1,700,000
FCTR at 31 Dec 20x4		Given	150,000
	2,000,000		2,400,000

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Problem 8.12

(1) Translation exposure at 31 December 20x3:

Maysub	(RM12,900) (net monetary liability)
Ozsub	A$18,280 (net assets)

(2) Functional currency of Maysub is the SGD (temporal method); functional currency of Ozsub is the Australian dollar (closing rate method)

Maysub's re-measured Profit and Loss Account
For the year ended 31 December 20x3

	RM	Rate	S$
Sales	60,000	0.4651	27,907
Cost of sales	(38,000)	Note (a)	(17,738)
Gross profit	22,000		10,169
Depreciation	(3,100)	Note (b)	(1,448)
Operating expenses	(9,900)	0.4651	(4,604)
Exchange loss	(1,000)	0.4545	(455)
Re-measurement gain (loss)		Note (c)	(90)
Net profit for the year	8,000		3,571

Note (a)

Cost of sales	RM	Exch. rate	S$	
Opening inventory	-		-	
Purchases	48,000	0.4651	22,326	
Closing inventory	(10,000)	0.4587	(4,587)	
	38,000			17,738

Note (b)

Depreciation expense (RM)	Exch. rate	Depreciation Expense (S$)
2,000	0.4695	939
1,100	0.4630	509
3,100		1,448

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Note (c)

Remeasurement difference:	RM	Exch. Rate	S$
Opening net monetary assets	20,000	0.4762	9,524
Sales	60,000	0.4651	27,907
Purchases of inventory	(48,000)	0.4651	(22,326)
Purchases of fixed assets:	(12,000)	0.4695	(5,634)
	(22,000)	0.4630	(10,185)
Operating expenses	(9,900)	0.4651	(4,604)
Exchange loss	(1,000)	0.4545	(455)
			(5,773)
Closing net monetary liabilities	(12,900)	0.4545	(5,863)
Re-measurement gain (loss)			90*

* rounding difference of $1

Maysub's remeasured Balance Sheet as at 31 December 20x3

	RM	Exch rate	S$
Fixed assets	30,900	Note d	14,371
Current assets:			
Inventory	10,000	0.4587	4,587
Accounts receivable	17,000	0.4545	7,727
Cash	6,100	0.4545	2,773
	33,100		15,087
Current liabilities			
Accounts payable	(12,000)	0.4545	(5,454)
Provisions	(2,000)	0.4545	(910)
	(14,000)	0.4545	(6,364)
Long-term loan	(22,000)	0.4545	(10,000)
Net assets	28,000		13,095
Share capital	20,000	0.4762	9,524
Retained earnings	8,000		3,571
	28,000		13,095

Note (d)

Fixed assets cost (RM)	Accum. Dep.	NBV	Exch. rate	S$
12,000	(2,000)	10,000	0.4695	4,695
22,000	(1,100)	20,900	0.463	9,676
34,000		30,900		14,371

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Ozsub's translated Income Statement
For the year ended 31 December 20x3

	A$000	Exch. Rate	S$000
Sales	35,000	1.15	40,250
Cost of sales	(24,750)	1.15	(28,462)
Gross profit	10,250	1.15	11,788
Depreciation	(2,000)	1.15	(2,300)
Amortisation exp	(500)	1.15	(575)
Operating expenses	(3,420)	1.15	(3,933)
Net profit for the year	4,330		4,980

Ozsub's translated Balance Sheet
As at 31 December 20x3

	A$'000	Rate	S$'000
Fixed assets	16,200	1.18	19,116
Patent	2,000	1.18	2,360
Current assets:			
Inventory	2,150	1.18	2,537
Prepayment	50	1.18	59
Accounts receivable	2,800	1.18	3,304
Cash	180	1.18	212
	5,180		6,112
Total assets	23,380		27,588
Current liabilities:			
Accounts payable	2,300	1.18	2,714
Deposits	800	1.18	944
Long-term loan	2,000	1.18	2,360
Share capital	10,000	1	10,000
Retained earnings	8,280	Note (a)	9,052
FCTR		Note (b)	2,518
Total liabilities and equity	23,380		27,588

Note (a):

Retained earnings	A$'000	Exch. rate	S$'000
31.12.20x1	1,500	1	1,500
31.12.20x2	2,450	1.05	2,573
31.12.20x3	4,330	1.15	4,980
	8,280		9,052

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Note (b)		A$000	Rate	S$000
	Opening net assets	11,500	1	11,500
	Net profit for year	2,450	1.05	2,573
				14,073
		13,950	1.1	15,345
	Translation difference 20x2			1,273
	Opening net assets	13,950	1.1	15,345
	Net profit for year	4,330	1.15	4,980
				20,325
	Closing net assets	18,280	1.18	21,570
	Translation difference			1,245
	FCTR, 1 January			1,273
	FCTR, 31 December			2,518

Problem 8.13

Functional currency of Maysub is the RM; functional currency of Ozsub is SGD.

Maysub's translated Income Statement
For the year ended 31 December 20x3

	RM	Rate	S$
Sales	60,000	0.4651	27,907
Cost of sales	(38,000)	0.4651	(17,674)
Gross profit	22,000		10,233
Depreciation	(3,100)	0.4651	(1,442)
Operating expenses	(9,900)	0.4651	(4,604)
Exchange loss	(1,000)	0.4545	(455)
Net profit for the year	8,000	0.4651	3,732

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Maysub's translated Balance Sheet as at 31 December 20x3

	RM	Rate	S$
Fixed assets	30,900	0.4545	14,045
Current assets:			
Inventory	10,000	0.4545	4,545
Accounts receivable	17,000	0.4545	7,727
Cash	6,100	0.4545	2,772
	33,100		15,045
Total assets	64,000	0.4545	29,089
Current liabilities			
Accounts payable	12,000	0.4545	5,454
Provisions	2,000	0.4545	909
	14,000		6,363
Long-term loan	22,000	0.4545	9,999
Share capital	20,000	0.4762	9,524
Retained earnings	8,000		3,732
FCTR		Note (a)	(529)
	64,000		29,089

Note (a)	RM	Exch. Rate	S$
Opening net assets	20,000	0.4762	9,524
Net profit for year	8,000	0.4651	3,732
			13,256
Closing net assets	28,000	0.4545	12,727
Translation difference			(529)*

* Rounding difference of 1

Ozsub's re-measured Profit and Loss Account
For the year ended 31 December 20x3

	A$'000	Rate	S$'000
Sales	35,000	1.15	40,250
Cost of sales	(24,750)	Note a	(28,279)
Gross profit	10,250		11,971
Depreciation	(2,000)	Note b	(2,024)
Amortisation exp	(500)	1.1	(550)
Operating expenses	(3,420)	1.15	(3,933)
Re-measurement diff		Note c	(401)
Net profit for the year	4,330		5,063
Retained earnings b/f	3,950	Note d	3,817
	8,280		8,880

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Note (a)	Cost of sales	A$'000		S$'000
	Opening inventory	2,000	1.08	2,160
	Purchases	24,900	1.15	28,635
	Closing inventory	(2,150)	1.17	(2,516)
		24,600		28,279

Note (b)

	A$'000	Rate	S$'000
Depreciation expense	800	1.00	800
Depreciation expense	1,200	1.02	1,224
Depreciation expense	2,000		2,024

Note (c)

	A$'000	Exch. rate	S$'000
Opening net monetary liabilities	(7,650)	1.1	(8,415)
Sales	35,000	1.15	40,250
Purchases of inventory	(24,900)	1.15	(28,635)
Purchases of fixed asset	(1,100)	1.19	(1,309)
Prepayment	(50)	1.18	(59)
Operating exp	(3,420)	1.15	(3,933)
			(2,101)
Ending net monetary liabilities	(2,120)	1.18	(2,502)
Remeasurement loss			(401)

Note (d)

The opening retained earnings cannot be translated directly as it is made up of pre-acquisition retained earnings and post-acquisition retained earnings. To obtain the post-acquisition retained earnings brought forward, it is necessary to reconstruct the opening balance sheet as follows:

Reconstructed balance sheet at 1 January 20x3

	A$	Rate	S$
Fixed assets	17,100	Note (e)	17,322
Patent	2,500	1.10	2,750
Inventory	2,000	1.08	2,160
Monetary items	(7,650)	1.10	(8,415)
	13,950		13,817
Share capital	10,000	1	10,000
Pre-acq. RE	1,500	1	1,500
Post-acq. RE	2,450	bal fig.	2,317
	13,950		13,817

The post-acquisition retained earnings is derived as a balancing figure.

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(e) Fixed assets at 1 Jan 20x3

Date acquired	Net book value	Rate	Re-measured amount
1/7/20x0*	6,000,000	1.0	6,000,000
31/3/20x2	11,100,000	1.02	11,322,000
	17,100,000		17,322,000

* Deemed as acquired on 31 December 20x1, the date Ozsub was acquired.

Ozsub
Translated balance sheet
As at 31 December 20x3

	A$'000	Exch. Rate	S$'000
Fixed assets	16,200	Note (f)	16,607
Patent	2,000	1.1	2,200
Current assets:			-
Inventory	2,150	1.17	2,516
Prepayment	50	1.18	59
Accounts receivable	2,800	1.18	3,304
Cash	180	1.18	212
	5,180		6,091
Total assets	23,380		24,898
Current liabilities			
Accounts payable	2,300	1.18	2,714
Unearned revenue	800	1.18	944
	3,100		3,658
Long-term loan	2,000	1.18	2,360
Share capital	10,000	1	10,000
Retained earnings	8,280		8,880
Total liabilities and equity	23,380		24,898

Note (f): Fixed assets

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Date acquired	Net book value (A$)	Rate	Remeasured amount (S$)
1/7/20x0*	5,200,000	1.0	5,200,000
31/3/20x2	9,900,000	1.02	10,098,000
25/12/20x3	1,100,000	1.19	1,309,000
	16,200,000		16,607,000

* Deemed as acquired on 31 December 20x1, the date Ozsub was acquired.

Problem 8.14

In this problem, in the absence of the fair value of NCI, assume the alternative accounting policy that non-controlling interests (NCI) is at the fair value of identifiable net assets, i.e. goodwill relating to NCI is not recognized.

The following calculations are required:

Goodwill

	$'000	LC'000
Cost of acquisition	285,000	285,000
Net assets @ fair value*	320,000	320,000
Less deferred tax on fair value differential**	(10,000)	(10,000)
Adjusted fair value of net assets	310,000	310,000
75% share	(232,500)	(232,500)
Goodwill	52,500	52,500**

* Share capital (120,000) + retained earnings (150,000) + differential (50,000)
**50,000 x 20%
***Goodwill is treated as an asset of the foreign operation.

Translation differences

On net assets:

	LC	Exch. Rate	$
Net assets (at book value) b/f	270,000	1.00	270,000
Net profit for 20x5	50,000	1.05	47,619
			317,619
Net assets (at book value) c/f	320,000	1.10	290,909
Translation loss			(26,710)

On fair value differential:

	LC	Exch. rate	$
Fair value differential b/f	50,000	1.00	50,000
Depreciation for 20x5	(10,000)	1.05	(9,524)
			40,476
Fair value differential c/f	40,000	1.10	36,364
Translation loss			(4,112)

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On Deferred tax on fair value differential

	LC	Exch. Rate	$
DTL, 1 January	10,000	1.00	10,000
Change in 20x5	(2,000)	1.05	(1,905)
			8,095
DTL, 31 December	8,000	1.10	7,273
Translation difference			822

On goodwill:

	LC	Exch. rate	$
Goodwill, 1 Jan	52,500	1.00	52,500
Goodwill, 31 Dec	52,500	1.10	47,728
Translation difference			(4,772)

The following CJEs are required:

CJE 1 (Figures are in $'000)

Dr	Share capital	120,000	
Dr	Retained Earnings	150,000	
Dr	Fixed assets	50,000	
Dr	Goodwill	52,500	
Cr	Deferred tax liability		10,000
Cr	Investment in R		285,000
Cr	Non-controlling interests		77,500

[Note: Deferred tax on fair value differential has been ignored.]

CJE 2

Dr	Depreciation	9,524	
Cr	Accumulated depreciation		9,524

[LC10,000 ÷ 1.05]
(Depreciation on undervalued assets)

CJE 2a

Dr	DTL	1,905	
Cr	Tax expense		1,905

(Tax effect on depreciation expense)

CJE 3

Dr	Translation loss on goodwill	4,772	
Cr	Goodwill		4,772

(Translation loss on goodwill)

CJE 4

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Dr	Translation loss on fair value differential	2,468		
Dr	NCI	822		
Dr	DTL	822		
Cr	Non-current assets			4,112

(Allocation of translation loss on fair value differential)

CJE 5

Dr	NCI (P/L)	10,000		
Cr	NCI (B/S)		10,000	

(NCI's share of net profit calculated as follows:

Net profit of Rockford (in $000)	47,619
Less depreciation of FV differential, after-tax	(7,619) [9,524 x 0.8]
	40,000
NCI's share @ 25%	10,000

CJE 6

Dr	NCI (B/S)	6,678		
CR	FCTR		6,678	

(NCI's share of FCTR: 25% of -$26,710)

CJE 7

Dr	Interest expense (Rockford)	1,800		
Cr	Interest income (Magnum)		1,800	

(Eliminate intercompany interest expense/income. It is assumed that interest income have been included under general and administrative expense as a credit item in Magnum's income statement.)

CJE 8

Dr	Loan payable	30,000		
Cr	Loan receivable		30,000	

(Eliminate intercompany loan)

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Rockford Company

Translated Income Statement for the year ended 31 December 20x2

	LC	Exch. Rate	$
Sales	350,000	1.05	333,333
Cost of goods sold	(240,000)	1.05	(228,571)
Gross profit	110,000	1.05	104,762
General and administrative expenses	(30,000)	1.05	(28,571)
Depreciation expense	(15,000)	1.05	(14,286)
Exchange loss	(3,000)	1.05*	(2,857)
Profit before tax	62,000		59,048
Taxation	(12,000)	1.05	(11,429)
Profit after tax	50,000	1.05	47,619
Retained earnings, 1 Jan	150,000	1	150,000
Retained earnings, 31 Dec	200,000		197,619

*exchange loss is translated at average rate. It is also acceptable to translate at actual rate.

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Rockford Company
Translated balance sheet
as at 31 December 20x2

	LC	Exch. Rate	$
Non-current assets	380,000	1.1	345,455
Current assets:			
Inventory	30,000	1.1	27,273
Accounts receivable	60,000	1.1	54,545
Cash	33,000	1.1	30,000
Total assets	503,000		457,273
Current liabilities	150,000	1.1	136,364
Non-current liabilities	33,000	1.1	30,000
Share capital	120,000	1	120,000
Retained earnings	200,000		197,619
FCTR			(26,710)
	503,000		457,273

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Magnum Group

Consolidated Financial Statements
For the year ended 31 December 20x5

	Magnum $	Rockford $	Consolidation Dr	Adjustments Cr	**Consolidated Income** Statement $
Sales	500,000	333,333			833,333
Cost of goods sold	(300,000)	(228,571)			(528,571)
Gross profit	200,000	104,762			304,762
General and administrative expenses	(75,000)	(28,571)	1,800	1,800	(103,571)
Depreciation expense	(50,000)	(14,286)	9,524		(73,810)
Exchange loss	-	(2,857)			(2,857)
Profit before tax	75,000	59,048			124,524
Taxation	(15,000)	(11,429)		1,905	(24,524)
Profit after tax	60,000	47,619			100,000
NCI's share of net profit			10,000		(10,000)
Retained earnings, 1 Jan	463,000	150,000	150,000		463,000
Retained earnings, 31 Dec	523,000	197,619	171,324	3,705	553,000

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	Magnum	Rockford	Consolidation Dr	Adjustments Cr	**Consolidated Balance sheet**
Goodwill			52,500	4,772	47,728
Non-current assets	750,000	345,455	50,000	9,524	1,131,819
				4,112	
Investment in Rockford	285,000			285,000	
Loan to Rockford	30,000			30,000	
Current assets:					
Inventory	150,000	27,273			177,273
Accounts receivable	80,000	54,545			134,545
Cash	18,000	30,000			48,000
Total assets	1,313,000	457,273			1,539,365
Current liabilities	210,000	136,364			346,364
Non-current liabilities	80,000	30,000	30,000		80,000
DTL			1,905		
			822	10,000	7,273
Share capital	500,000	120,000	120,000		500,000
Retained earnings	523,000	197,619	171,324	3,705	553,000
					-
FCTR		(26,710)	4,772	6,678	(27,272)
			2,468		
NCI			822	77,500	80,000
			6,678	10,000	
	1,313,000	457,273	441,291	441,291	1,539,365

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Problem 8.15

Calculation of gain from sale of associate

(1) **At company level**:

Sales proceeds [FC18,000,000 x 1.23]	22,140,000
Cost of investment (FC10,000,000 x 1.2)	(12,000,000)
Gain on sale	10,140,000

At group level:

Sales proceeds [18,000,000 x 1.23]	22,140,000
Carrying value*	(18,450,000)
Gain on sale	3,690,000
Add: FCTR transferred to I/S	110,000
Total gain reported in consolidated I/S	3,800,000

*Carrying value:

Cost of investment	12,000,000	
Retained earnings	6,340,000	(Note a)
FCTR	110,000	(Note a)
	18,450,000	

Reconciliation check:

Gain on sale in parent's books	10,140,000
Less cumulative profit recognized at Group level	(6,340,000)
= Gain reported in group accounts	3,800,000

Note (a): FCTR

20x1	LC	Exch. Rate	$
Net assets (at book value) b/f	10,000	1.20	12,000
Net profit for 20x5	2,000	1.25	2,500
			14,500
Net assets (at book value) c/f	12,000	1.30	15,600
Translation loss			1,100

20x2	LC	Exch. Rate	$
Net assets (at book value) b/f	12,000	1.30	15,600
Net profit for 20x5	3,000	1.28	3,840
			19,440
Net assets (at book value) c/f	15,000	1.23	18,450
Translation loss			(990)
Cumulative FCTR			110

Retained earnings at 31.12.20x2 = $2,500 + $3,840 =$6,340

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Problem 8.16
(1)

- Co Y is a 80% owned subsidiary of SingCo. The functional currency of Co Y is the SGD.
- The financial statements kept in SGD are shown below.
- Translate the financial statements into SGD the presentation currency.

Income Statement of Co Y for the year ended 31 Dec 2012

		SGD	Rate	USD
Sales		2,000,000	0.595	1,190,000
Cost of Sales				
Opening inventory	320,000			
Purchases	1,500,000			
	1,820,000			
Less closing inventory	(480,000)			
Less Cost of Sales		(1,340,000)	0.595	(797,300)
Gross profit		660,000		392,700
Operating expenses		(100,000)	0.595	(59,000)
Interest expense		(16,000)	0.595	(9,520)
Depreciation		(80,000)	0.595	(47,600)
Profit before tax		464,000		276,080
Tax expense		(120,000)	0.595	(71,400)
Profit after tax		344,000		204,680
Dividends declared		(40,000)	0.606	(24,240)
Profit retained		304,000		180,440
Retained earnings, 1 Jan 20x5		400,000	Note 1	262,000
Retained earnings, 31 Dec 20x5		704,000	To BS	442,440

Statement of Financial Position as at 31 Dec 2012

		USD	Rate	SGD
Fixed assets, net book value		640,000	0.625	400,000
Inventory		480,000	0.625	300.000
Accounts receivable		200,000	0.625	125,000
Cash		20,000	0.625	12,500
		1,340,000		837,500
Accounts payable		436,000	0.625	272,500
Share capital		200,000	0.476	95,200
Retained earnings				
Pre-acquisition:	240,000		0.476	114,240
Post-acquisition:	464,000			328,200
		704,000	From I/S	442,440
Foreign Currency Translation Reserve (FCTR)				27,360
		1,340,000		837,500

Note 1: Reconstruct the Statement of Financial Position as at 31 Dec 2011 to determine the RE in SGD at the beginning of the year

	USD	Rate	SGD

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Statement of Financial Position as at 31 Dec 2011			
Net assets	600,000	0.562	337,200
Share capital	200,000	0.476	95,200
Retained earnings	400,000		262,000
FCTR			(20,000)
	600,000		337,200

(2)

	SGD	Rate	USD
Determination of goodwill			
Consideration transferred	600,000		
Fair value of NCI	60,000		
Less fair value of identifiable net assets			
Share capital	200,000		
Pre-acquisition retained earnings	240,000		
Add under-valued inventory	40,000		
Less deferred tax liability on under-valuation	(8,000)		
Goodwill at acquisition date	188,000	0.476	89,488
Translation gain			28,012
Goodwill at 31 December 2012	188,000	0.625	117,500
Under-valued inventory at acquisition date	40,000	0.476	19,040
Translation gain			5,960
Under-valued inventory at 31 December 2012	40,000	0.625	25,000
Deferred tax liability on under-valuation at acquisition date	(8,000)	0.476	(3,808)
Translation loss			(1,192)
Deferred tax liability on under-valuation at 31 December 2012	(8,000)	0.625	(5,000)

(3)

Figures in USD		
Dr Share capital	95,200	
Dr Retained earnings	114,240	
Dr Inventory	19,040	
Dr Goodwill	89,488	
Cr Deferred tax liability		3,808
Cr Investment		285,600
Cr Non-controlling interests		28,560
Elimination of investment		
Dr Goodwill	28,012	
Dr Inventory	5,960	
Cr Deferred tax liability		1,192
Cr FCTR		32,780

Translation gain/loss on goodwill, under-valued inventory, deferred tax liability on under-valued inventory

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Dr RE (20% x ($262,000 - $114,240))	29,552	
Cr NCI		29,552
Allocation of post-acquisition RE to NCI		
Dr Income to NCI (20% x $204,680)	40,936	
Cr NCI		40,936
Allocation of income to NCI		
Dr Dividend income	19,392	
Dr NCI	4,848	
Cr Dividend declared		24,240
Elimination of dividend declared		
Dr FCTR	6,426	
Cr NCI		6,426
Allocation of FCTR to NCI		

Translation gain on net book value (27,360 x 20%) = $5,472
Translation gain/loss on goodwill = 0
Goodwill on NCI is zero as it cannot be negative
(60,000 – 20% x (200,000 + 240,000 + 40,000 – 8,000))
Translation gain on inventory/deferred tax liability
= (40,000 – 8,000) x 20% x (0.625 – 0.476) = 954

Problem 8.17

	15 September 2014		31 December 2013	
Income statement	USD	SGD	USD	SGD
Sales	245,000	330,750	420,000	525,000
Expenses	(189,000)	(255,150)	(296,000)	(370,000)
Net profit before tax	56,000	75,600	124,000	155,000
Statement of financial position				
Fixed assets	490,000	695,800	432,000	561,600
Inventory	165,000	234,300	129,780	168,714
Accounts receivable	120,000	170,400	148,900	193,570
Bank	32,000	45,440	45,600	59,280
	807,000	1,145,940	756,280	983,164
Accounts payable	187,000	265,540	232,280	301,964
Share capital	300,000	426,000	300,000	360,000
Retained earnings	180,000	255,600	124,000	155,000
OCI	140,000	198,800	100,000	130,000
FCTR				36,200
	807,000	1,145,940	756,280	983,164
Statement of changes in equity				
Share capital				
Balance, 1 Jan	300,000	360,000	300,000	360,000
Translation adjustments		66,000		
Balance, 31 Dec	300,000	426,000	300,000	360,000

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Retained earnings				
Balance, 1 Jan	124,000	155,000	0	0
Net profit after tax	56,000	75,600	124,000	155,000
Translation adjustments		25,000	0	0
Balance, 31 Dec	180,000	255,600	124,000	155,000
OCI				
Balance, 1 Jan	100,000	130,000	0	0
Revaluations	40,000	56,000	100,000	130,000
Translation adjustments		12,800	0	0
Balance, 31 Dec	140,000	198,800	100,000	130,000
FCTR				
Balance, 1 Jan		36,200		
Translation adjustments during year				
- Opening net assets		524000x(1.42-1.3)= 62,880		
- Income during year		56000x(1.42-1.35) = 3,920		
- OCI change during year		40000x(1.42-1.4) = 800		
Effect of change in functional currency				
- Adjustment of FCTR to share capital		(66,000)		
- Adjustment of FCTR to retained earnings		(25,000)		
- Adjustment of FCTR to OCI		(12,800)		
Balance, 31 Dec		0		

Problem 8.18 Exchange rate effects on loss of control of foreign subsidiary

Translate the financial position and determine the foreign currency translation reserve of the Singapore subsidiary at the date of disposal on 31 December 20x6

Balance Sheet at 31 December 20x6	S$'000	Rate	US$'000	
Other assets	30,800	1.68	18,333	
Other liabilities	(4,800)	1.68	(2,858)	
	26,000		15,475	
Represented by:				
Share capital	200	1.55	129	
Retained earnings	25,800	*	15,374	
Foreign currency translation reserve	-		(28)	(Balancing)
	26,000		15,475	

* At the respective average rates. Please refer to the computation under analytical check.
Analytical check on the foreign currency translation reserve at the date of disposal

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		S$'000	Transaction Rate	US$'000	Closing Rate	US$'000	Difference
		(1)	(2)	(3)=(1)/(2)	(4)	(5)=(1)/(4)	(6)=(5)-(3)
	Share capital	200	1.55	129	1.68	119	(10)
	Retained earnings						
20x3	Profit for the year	2,400	1.74	1,379	1.68	1,428	
20x4	Profit for the year	6,200	1.69	3,669	1.68	3,690	
20x5	Profit for the year	7,500	1.66	4,518	1.68	4,464	
20x6	Profit for the year	9,700	1.67	5,808	1.68	5,774	(18)
		26,000		15,503		15,475	
					Cumulative exchange difference		(28)

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

	US$'000
Sales proceeds	8,000
Less: Cost of investment in separate financial statements	(90)
Gain on disposal	7,910

In the consolidated financial statements

	US$'000
Sales proceeds	8,000
Less: Carrying value of net assets at the date of disposal	(15,475)
Less: Foreign currency translation reserves	(28)
Add: Fair value of retained interest	10,500
Gain on disposal	2,997

Explanatory notes

1. Gain on disposal on the consolidated level is effectively as follows:

Gain on disposal at consolidated level	=	Gain on disposal in separate financial statements	-	Post-acquisition profits of Singapore subsidiary (at transaction rates) & remaining 30% share capital	+	Retained interest of 30% at fair value
	=	US$ 7,910,000	-	US$15,374,000 – US$39,000	+	US$10,500,000
	=	US$2,997,000				

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Prepare the accounting entries
In the separate financial statements

Dr	Cash	8,000,000	
Cr	Cost of investment in subsidiary		90,000
Cr	Gain on disposal of investment		7,910,000

(Being disposal of investment in subsidiary in the separate financial statements)

In the consolidated financial statements

Dr	Gain on disposal	4,913,000	
Dr	Retained interest of 30% (FV)	10,500,000	
Cr	Beginning retained profits		15,374,000
Cr	Cost of investment in subsidiary		39,000

(Being adjustment of gain on disposal of subsidiary at group level)

Problem 8.19 Exchange rate effects on partial disposal of foreign subsidiary

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

	US$'000
Sales proceeds	8,000
Less: Cost of investment in separate financial statements	(13)[+]
Gain on disposal	7,987

[+] Derived as 10%*129,000

In the consolidated financial statements

	US$'000	
Sales proceeds	8,000	
Less: Proportionate share of net assets	(1,548)	[10%*15,475,000]
Less: Proportionate share of foreign currency translation reserves	(3)	[10%*28,000]
Premium on disposal of interests to NCI (Note 1)	6,449	

Explanatory notes

1. As the US parent company still retains control, the partial disposal is accounted for as an equity transaction i.e. transactions with owners in their capacity as owners in the consolidated financial statements. Accordingly, the difference between the amount by which the non-controlling interests are adjusted and the fair value of the consideration received is recognised in equity and attributed to the owners of the US parent in accordance with B96 of IFRS 10.

2. Gain on disposal on the consolidated level is effectively as follows:

Gain on disposal at consolidated level	=	Gain on disposal in separate financial statements	-	Parent's disposed share of post-acquisition profits of Singapore subsidiary (at average rates)
	=	7,987,000	-	10%*15,374,000

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= 7,987,000 - 1,538,000
= 6,449,000

Calculate the relative proportion of equity items to be allocated to non-controlling interests

		US$'000	**Non-controlling interests (10%)** US$'000	**Parent's interest (90%)** US$'000
	Share capital	129	13	116
	Revenue reserves			
20x3	Profit for the year	1,379	138	1,241
20x4	Profit for the year	3,669	367	3,302
20x5	Profit for the year	4,518	452	4,066
20x6	Profit for the year	5,808	581	5,227
		15,374	1,538	13,836
	Foreign currency translation reserve	(28)	(3)	(25)
		15,447	1,544	13,903

Prepare accounting entries

In the separate financial statements

Dr	Cash	8,000,000	
Cr	Cost of investment in subsidiary		13,000
Cr	Gain on disposal		7,987,000

(Being disposal of 10% interests in investment in subsidiary in the separate financial statements)

In the consolidated financial statements

Dr	Share capital	116,000	
Cr	Investment in subsidiary		116,000 [129,000 – 13,000]

(Being elimination of parent's investment in subsidiary and share capital as well as pre-acquisition reserves of subsidiary)

Dr	Gain on disposal of investment	7,987,000	
Cr	Premium on disposal of interests to NCI		7,987,000

(Being reclassification of gain on disposal of investment to premium on disposal of interest to NCI)

Dr	Premium on disposal of interests to non-controlling interests	1,538,000	
Dr	Share capital	13,000	
Cr	Foreign currency translation reserve		3,000
Cr	Non-controlling interests		1,544,000

(Being reattribution of other comprehensive income and elimination of share capital attributable to non-controlling interests)

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Problem 8.20 Exchange rate effects on disposal of foreign joint ventures

Calculation and translate the goodwill on acquisition of associate

Date of acquisition based on US parent's 35% interest

	S$'000	Transaction Rate	US$'000
Consideration paid	800	1.75	457
Less: Fair value of net identifiable assets	(710)	1.75	(406)
Goodwill	90		51

Translation of goodwill

		S$'000	Closing rate	US$'000
20x2	Balance at beginning and end	90	1.75	51
	Exchange differences	-		4
20x5	Balance at date of disposal	90	1.64	55

Translate the financial position and determine the parent's share of foreign currency translation reserve of the Singapore associate at the date of disposal on 31 December 20x5

Balance Sheet at 31 December 20x5	S$'000	Closing Rate	US$'000	Parent's 35% Interest
Other assets	86,600	1.64	52,805	18,482
Other liabilities	(5,800)	1.64	(3,537)	(1,238)
	80,800		49,268	17,244
Represented by:				
Share capital	800	1.75	457	160
Retained earnings	80,000	*	47,346	16,571
Foreign currency translation reserve	-		1,465	513
	80,800		49,268	17,244

* At the respective average rates. Please refer to computation under analytical check.

	US$'000
Parent's share of foreign currency translation reserve	513
Add: Exchange differences on translation of goodwill	4
Parent's share of total foreign currency translation reserve	517

Analytical check on the total foreign currency translation reserve at the date of disposal before accounting for parent's share of exchange difference on translation of goodwill

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		S$'000	Transaction Rate	US$'000	Closing Rate	US$'000	Difference
		(1)	(2)	(3)=(1)/(2)	(4)	(5)=(1)/(4)	(6)=(4)-(2)
	Share capital	800	1.75	457	1.64	488	31
	Retained earnings						
20x2	Balance b/f	3,300	1.75	1,886	1.64	2,012	
20x3	Profit for the year	18,200	1.74	10,459	1.64	11,098	
20x4	Profit for the year	22,500	1.69	13,314	1.64	13,719	
20x5	Profit for the year	36,000	1.66	21,687	1.64	21,951	
		80,000		47,346		48,780	1,434
	Cumulative exchange difference before exchange difference on translation of goodwill						1,465

Calculate the gain on disposal in the separate and consolidated financial statements

In the separate financial statements

	US$'000
Sales proceeds	4,000
Less: Cost of investment in separate financial statements	(196)^
Gain on disposal	3,804

^ Derived as 0.15/0.35*carrying value of US$457,000

In the consolidated financial statements

	US$'000
Sales proceeds	4,000
Less: Carrying value of net assets (Note 1)	(7,236)
Add: Foreign currency translation reserves	221
Loss on disposal (Note 2)	(3,015)

Explanatory notes

1. The carrying value of net assets is derived as:

Investment in associate	US$'000	
Cost	457	
Share of beginning retained profits	8,321	[35%*(10,459,000+13,314,000)]
Share of profits for the year	7,590	[35%*21,687,000]
Share of foreign currency translation reserve	516	(inclusive of exchange difference of translation of goodwill)
	16,884	
Analytical check:		
Share of carrying value of associate	17,244	[35%*49,268,000]
Less: Excess of carrying value over FV	(414)	[1,435,000-710,000]/1.75

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Parent's share of goodwill	54
	16,884
Therefore:	
15% disposed	7,236

2. Loss on disposal at the consolidated financial statements level is effectively as follows:

Loss on disposal at consolidated financial statements level	=	Gain on disposal in separate financial statements	-	Parent's disposed share of post-acquisition profits of Singapore associate (at average rates)
	=	3,804,000	-	[(0.15/0.35)*(7,590,000+8,321,000)
	=	3,804,000	-	6,819,000
	=	3,015,000		

Prepare the accounting entries

In the separate financial statements

Dr	Cash	4,000,000	
Cr	Cost of investment in associate		196,000
Cr	Gain on disposal		3,804,000

(Being disposal of 15% interests in investment in associate in the separate financial statements)

In the consolidated financial statements

Dr	Investment in associate	16,424,000	
Cr	Share of beginning retained profits		8,321,000
Cr	Share of associate's profit for the year		7,590,000
Cr	Share of associate's foreign currency translation reserve		513,000

(Being equity accounting for associate's share of results up to the date of disposal)

Dr	Investment in associate	4,000	
Cr	Share of associate's foreign currency translation reserve		4,000

(Being parent's share of exchange difference of translation of goodwill)

Dr	Gain on disposal	6,819,000	
Dr	Share of associate's foreign currency translation reserve	221,000	
Cr	Investment in associate		7,040,000

(Being adjustment for gain on disposal at consolidated financial statements level)

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